

Physical Human–Robot Interaction of a Robotic Exoskeleton By Admittance Control

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Abstract—In this paper, a physical human–robot interaction approach is presented for the developed robotic exoskeleton using admittance control to deal with a human subject's intention as well as the unknown inertia masses and moments in the robotic dynamics. The human subject's intention is represented by the reference trajectory when the robotic exoskeleton is complying with the external interaction force. Online estimation of the stiffness is employed to deal with the variable impedance property of the robotic exoskeleton. Admittance control is first presented based on the measured force in order to generate a reference trajectory in interaction tasks. Then, adaptive control is proposed to deal with the uncertain robotic dynamics and a stability criterion can be obtained. Bounded errors are shown in the motion tracking while the robustness of the variable stiffness control is guaranteed. The experimental results indicate that the proposed control enables the human subjects to execute an admittance control task on the exoskeleton robot effectively.

Index Terms—Admittance control, human–robot interaction, robotic exoskeleton, variable stiffness.

I. INTRODUCTION

PHYSICAL human–robot interaction has been widely applied in many areas, such as rehabilitation therapies and cooperative manipulation tasks. In order to perform a given interaction task, for example, a point-to-point drawing motion was considered in [1], impedance control or admittance control methods are usually utilized to deal with such issues. In the human–robot system, the control of assistive robots aims

at minimizing the workload for humans to accomplish the given tasks, such that the human efforts can be reduced in the execution. In order to address the issue, this paper shall propose a method to deal with physical human–robot interaction (pHRI) with minimum interaction force as well as task tracking errors.

Admittance control accepts a force as input and reacts with the robot's motions. The appropriate choice of the mass, damper, and spring coefficients can make the admittance control conform to the required effect. In order to deal with the instability issues generated by force control, the concept of impedance control was proposed in [2], and [3]. But if impedance control is to be performed, the robotic dynamics and the interaction dynamics are required [4], [5]. As the dual form of the above approach, admittance control has also been widely used in the applications of pHRI. In [6], admittance control with virtual force was presented to make the robot perform accurately in dancing state, which was an extreme application in pHRI. In [7], the motions of robots were generated by the external human force using admittance control. A framework containing an outer loop and an inner loop should be investigated, where the robot is able to track the output trajectory of the admittance model. However, the control should not contain a specific performance task model or a human dynamic model. In [8], an omnidirectional-type cane robot was described in a transfer function, but the virtual mass and damper coefficients were both constants and the authors neglected the stiffness coefficient in the admittance controller. In [9], admittance control was presented in pHRI, where the robotic model was described as a spring–mass–damper system with the three coefficients being constants.

Recently, pHRI is implemented in the controlling of robots with tactile sensing devices to achieve compliance between humans and robots by different forms of admittance control [10]. The adaptation or learning are investigated when humans learn to contact a robotic equipment, where it includes two parts. The first part contains learning a robotic model to make the compensation for the robot dynamics. In [11], the optimal parameters of the impedance were obtained from a natural algorithm in robot interaction tasks. In [12], an interaction task between robots and unknown steady environments was performed by a proposed adaptive impedance learning. The second part contains learning a control loop which validates the effect of the cooperative tasks that are related to a human–robot model. In [13], variable impedance control was performed in the task of minimizing an objective function, using reinforcement learning. The tasks of pHRI were fulfilled by using adaptive impedance control. These

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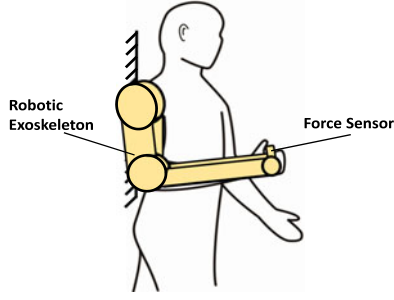


Fig. 1. Human–robot interaction task model.

works show that the controllers for such specific tasks should be closed in the outer loop, which contains human factors as well as a desired model for the performance. In order to perform human–robot interaction tasks more easily, some recent research works have paid attention to minimization of the effort for the operator. In [14], variability inertia was adapted to maintain the passivity of the overall system, but the spring item was not considered in the impedance system.

Additionally, reference trajectory adaption should also be taken into consideration in order to acquire the ideal adaption performance besides the impedance learning mentioned above [15]. Although the effect of the tracking performance was guaranteed as the control purpose, the interaction force was treated as an item of external disturbance and the compliance between the external force and robot was not taken into consideration. The field of pHRI has also involved reference trajectory adaption, where the movement of the human is obtained for the updation of the robot's trajectory, and then, the interaction would be compliant according to the motions of human and robot. In [16], the reference trajectory adaption was used in the collaborative tasks between humans and robots, where human characteristics of motions were taken into consideration. The intended movement of the human was estimated by the robot and used to control the manipulator while admittance control was to yield compliance with the interaction force. In [17], the reference trajectory shaping was presented only to make the robot governed by a given impedance model while the constraint satisfaction was guaranteed at the same time.

Based on above discussions, a framework of the adaptive admittance control with time-varying stiffness parameter is proposed, which is capable of dealing with the human's motion intention so that it can perform more accurately in the actual physical interaction. The control approach is able to apply to humans of different skill levels and variant force powers without prior offline model tuning, and the robustness of the controller is guaranteed. The control scheme consists of an inner loop and an outer loop. The former is able to deal with the unknown masses and moments of inertia in the robot dynamics, while the latter is to tune the interaction model considering the intention of human subjects. The interaction model is shown in Fig. 1 and the overall control framework is shown in Fig. 2. The contributions of this paper are listed as follows.

- 1) The reference trajectory of the robotic exoskeleton would be reshaped according to the human–robot interaction force.

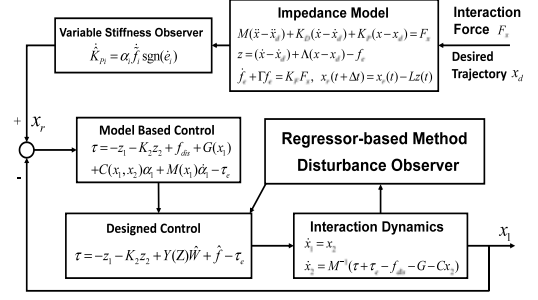


Fig. 2. Block diagram of the control scheme.

- 2) The human intention is considered and represented by the shaped reference trajectory from the outer loop.
- 3) The developed adaptive controller is proposed to deal with the uncertain nonlinear robotic dynamics.

II. PRELIMINARIES AND PROBLEM FORMULATION

The dynamics of an n -link robotic exoskeleton interacting with the human subject (the interaction model is shown in Fig. 1) can be described as follows:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) + f_{\text{dis}}(t) = \tau + \tau_e(t) \quad (1)$$

where $q \in \mathbb{R}^n$ is the position coordinates of the robotic joints, $\tau \in \mathbb{R}^n$ is the applied torque as the input item, $\tau_e \in \mathbb{R}^n$ is the torque in interaction tasks with the environment (or human), $M(q) \in \mathbb{R}^{n \times n}$ is the inertia matrix of symmetric positive definite of the robotic exoskeleton, $C(q, \dot{q}) \in \mathbb{R}^{n \times n}$ is considered as the centripetal and Coriolis torques, $G(q) \in \mathbb{R}^n$ is the force of gravity contributed by the robotic exoskeleton, and $f_{\text{dis}}(t) \in \mathbb{R}^n$ is considered as the external disturbance to the robot system. The terms $M(q)$, $C(q, \dot{q})$, and $G(q)$ include uncertain dynamic parameters.

Property 1 (see [22]): The matrix $\dot{M}(q) - 2C(q, \dot{q})$ is skew-symmetric.

Property 2 (see [23]): Its inverse $M^{-1}(q)$ exists, and is also positive definite and bounded, i.e., $\|M^{-1}(q)\| < \alpha_{M^{-1}}$, where $\alpha_{M^{-1}}$ is a positive constant.

Property 3 (see [24]): There exists a vector $W \in \mathbb{R}^m$ with components depending on the robotic exoskeleton parameters (masses, moments of inertia, etc.), such that for the exoskeleton dynamics (1), we have

$$M(q)\dot{\zeta} + C(q, \dot{q})\zeta + G(q) = Y(q, \dot{q}, \zeta, \dot{\zeta})W \quad (2)$$

where $Y(q, \dot{q}, \zeta, \dot{\zeta}) \in \mathbb{R}^{n \times m}$ is called the dynamic regressor matrix and $\zeta \in \mathbb{R}^m$ is a vector of differentiable function.

We can transform the joint space into task space and the relation yields $x = \Omega(q)$, $\dot{x} = J(q)\dot{q}$. Consider an exoskeleton with two joints, actually we have

$$\begin{cases} x = L_1 \cos(q_1) + L_2 \cos(q_1 + q_2) \\ y = L_1 \sin(q_1) + L_2 \sin(q_1 + q_2) \end{cases} \quad (3)$$

where L_1 and L_2 are the length of joint links, q_1 and q_2 are the joint angles, and x and y are the coordinates in the task space.

Then, we have the robot Jacobian

$$J(q) = \begin{bmatrix} -L_1 \sin(q_1) - L_2 \sin(q_1 + q_2) & -L_2 \sin(q_1 + q_2) \\ L_1 \cos(q_1) + L_2 \cos(q_1 + q_2) & L_2 \cos(q_1 + q_2) \end{bmatrix} \quad (4)$$

According to (1), let $x_1 = [q_1, q_2, q_3, \dots, q_n]^T$, $x_2 = [\dot{q}_1, \dot{q}_2, \dot{q}_3, \dots, \dot{q}_n]^T$, and $F_x = J^{-T} \tau_e$, where x_1 is the actual trajectory, x_2 is the actual velocity, and F_x is the interaction force. The purpose of the control is to make the joint variable x_1 keep tracking the reference path x_r while there exists an interaction force. In addition, the closed-loop signals are required to be bounded and converged as well as preventing the position constraints $|x_{1,i}(t)| < k_{c_i}$, $i = 1, 2, \dots, n$ from being violated $\forall t > 0$.

Assumption 1: Positive constants k_{d_i} , $i = 1, 2, \dots, n$, exist, such that $|x_{r_i}(t)| \leq k_{d_i} < k_{c_i}$, $i = 1, \dots, n$, $\forall t \geq 0$.

Assumption 2: A positive constant F_{x_m} exists, so we have $\|F_x(t)\| \leq F_{x_m} \forall t \geq 0$, where F_{x_m} denotes the maximum of the interaction force.

Assumption 3: For $\forall t \in R^+$, the unknown external disturbance f_{dis} and its derivative $\dot{f}_{\text{dis}}$ are bounded, i.e., $\|f_{\text{dis}}\| \leq f_M^*$, $\|\dot{f}_{\text{dis}}\| \leq f^*$, where f_M^* and f^* are positive constants.

Lemma 1 (see [21]): If a Lyapunov function $V(x)$ exists in the consideration of initially bounded, which is positive definite and C^1 continuous, and satisfies $\kappa_1(\|x\|) \leq V(x) \leq \kappa_2(\|x\|)$, such that $\dot{V}(x) \leq -\kappa V(x) + c$, where $\kappa_1, \kappa_2 : \mathbb{R}^n \rightarrow \mathbb{R}$ are class $\mathcal{K}$ functions, while both κ and c are positive constants, then the solution to the Lyapunov function, $x(t)$, is uniformly bounded.

III. ADAPTIVE ADMITTANCE CONTROL DESIGN

A. Reference Trajectory Shaping

While interacting with the human, the robotic exoskeleton will track a new trajectory that deviates from the desired trajectory, which is due to the human subject's intention. In general, the human intention can be represented by the new trajectory, namely reference trajectory. Considering the decrease in the interaction force and the convergence of the tracking errors in the meantime, we assume that the adaption to the desired trajectory is to minimize the following cost function:

$$\Phi = \int_0^T \|F_x\|_R + \|x(\delta) - x_d(\delta)\|_G d\delta \quad (5)$$

where $\|\cdot\|_R$ and $\|\cdot\|_G$ are norms of the matrix while R and G are weights. Then, there is a balance between the human force and the error of the reshaped trajectory.

An impedance model for the exoskeleton is used to solve the cost function (5), then we have

$$M(\ddot{x} - \ddot{x}_d) + D(\dot{x} - \dot{x}_d) + K(x - x_d) = F_x \quad (6)$$

where $x \in \mathbb{R}^n$ is the position of the exoskeleton joint, $x_d \in \mathbb{R}^n$ is the desired position, and M , D , K , and F_x are the inertia, damping, stiffness, and force matrix, respectively. The external force for interaction and the error in the tasks can also be regulated using the above model, and x_d is the initial desired

trajectory. x and F_x solved from (6) will minimize Φ in the cost function.

Equation (6) can be equivalently written as

$$w = (\ddot{x} - \ddot{x}_d) + K_D(\dot{x} - \dot{x}_d) + K_P(x - x_d) - K_F F_x \quad (7)$$

where w , x , x_d and its first and second derivatives are the functions of time t , $w \in \mathbb{R}^n$, $K_D \in \mathbb{R}^{n \times n}$, $K_P \in \mathbb{R}^{n \times n}$ and $K_F \in \mathbb{R}^{n \times n}$, $K_D = M^{-1}D = \text{diag}[K_{D1}, K_{D2}, \dots, K_{Dn}]$, $K_P = M^{-1}K = \text{diag}[K_{P1}, K_{P2}, \dots, K_{Pn}]$, $K_F = M^{-1} = \text{diag}[K_{F1}, K_{F2}, \dots, K_{Fn}]$.

Let $\ddot{y} = \ddot{x} - \ddot{x}_d$, $\dot{y} = \dot{x} - \dot{x}_d$, and $y = x - x_d$. We consider a mass-damper-spring system $f = \ddot{y} + K_D \dot{y} + K_P y$, where $f \in \mathbb{R}^n$ is the applied force and $f = [f_1, f_2, \dots, f_n]^T$. Now, we consider a force function $h(y, u)$, as defined in [18], taking place of the spring item, where the stiffness regulating input $u(t)$ and the derivative $\dot{u}(t)$ are both bounded. We can obtain

$$f_i = \ddot{y}_i + K_{Di} \dot{y}_i + h(y_i, u_i) \quad (8)$$

where $i = 1, 2, \dots, n$. Then, we assume that the ratio between $\dot{u}_i(t)$ and the velocity error of the robot joint is bounded, i.e., for any time t , the inequality holds $\frac{|\dot{u}_i(t)|}{|\dot{y}_i(t)|} < v \in \mathbb{R} \forall t$. The stiffness to be obtained is

$$\frac{\partial f_i}{\partial y_i} = \frac{\partial h(y_i, u_i)}{\partial y_i} = K_{Pi}(y_i, u_i(t)). \quad (9)$$

According to [18], the stiffness $K_{Pi}(y_i, u_i(t))$ is time-varying. Let $\hat{K}_{Pi}(t)$ be the estimated stiffness, and $\tilde{K}_{Pi}(t) = K_{Pi}(y_i, u_i(t)) - \hat{K}_{Pi}(t)$ represent the estimation error. Taking the differential form of (8) with respect to time

$$\dot{f}_i = \ddot{y}_i + K_{Di} \dot{y}_i + K_{Pi} \dot{y}_i + h_{ui} \dot{u}_i \quad (10)$$

where $h_{ui} = \frac{\partial h(y_i, u_i)}{\partial u_i}$. The estimate of $\dot{f}_i$ is as follows:

$$\hat{\dot{f}}_i = \ddot{y}_i + K_{Di} \dot{y}_i + \hat{K}_{Pi} \dot{y}_i. \quad (11)$$

The update law of the estimated stiffness is

$$\dot{\hat{K}}_{Pi} = \alpha_i \hat{f}_i \text{sgn}(\dot{y}_i) \quad (12)$$

with $\hat{\dot{f}}_i = \dot{f}_i - \hat{\dot{f}}_i$, $\alpha_i > 0$ and

$$\text{sgn}(\bullet) = \begin{cases} \frac{\dot{\bullet}}{\|\bullet\|}, & \|\bullet\| \neq 0 \\ 0, & \|\bullet\| = 0 \end{cases}. \quad (13)$$

In order to calculate the stiffness $\hat{K}_{Pi}$ from the updating law in (12), $\hat{\dot{f}}_i$ needs to be calculated first. Note that the item $\ddot{y}_i$ appears both in (10) and (11), thus for convenience, the item can be canceled out in the equation $\dot{f}_i - \hat{\dot{f}}_i$. Actually, $\dot{f}_i$ denotes the time derivative of the interaction force and can be obtained in the sampling of the force sensor. The item $\hat{\dot{f}}_i$ is calculated using the stiffness K_{Pi} , the damping K_{Di} , and in the absence of $\ddot{y}_i$ as is mentioned before. Thus, we can obtain $\hat{\dot{f}}_i = \dot{f}_i - \hat{\dot{f}}_i$ in the calculation of the estimated stiffness.

Consider the positive definite error function

$$V_k = \sum_{i=1}^n \frac{1}{2} \tilde{K}_{Pi}^2 \quad (14)$$

and its derivative with respect to time, we can obtain

$$\begin{aligned}\dot{V}_k &= \sum_{i=1}^n (\tilde{K}_{Pi} \dot{K}_{Pi} - \alpha_i \tilde{K}_{Pi} h_{ui} \dot{u}_i \operatorname{sgn}(\dot{y}_i) - \alpha_i \tilde{K}_{Pi}^2 |\dot{y}_i|) \\ &\leq \sum_{i=1}^n (\tilde{K}_{Pi} \dot{K}_{Pi} - \alpha_i \tilde{K}_{Pi} h_{ui} \dot{u}_i \operatorname{sgn}(\dot{y}_i)).\end{aligned}\quad (15)$$

According to [18], when the inequality holds $|\tilde{K}_{Pi}| > |h_{ui}| \frac{|\dot{u}_i|}{|\dot{y}_i|} + \frac{1}{\alpha_i} \frac{|\dot{K}_{Pi}|}{|\dot{y}_i|}$, one can obtain $\dot{V}_k < 0$, thus the estimation error of the stiffness would decrease. By writing $\dot{K}_{Pi} = K_{yi} \dot{y} + K_{ui} \dot{u}$ and consider the upper bound assumed above, i.e., $\frac{|\dot{u}_i(t)|}{|\dot{y}_i(t)|} < v \in \mathbb{R} \forall t$, the estimation of the stiffness would converge to the true value, and the ultimate uniform error is given in the following form:

$$|\tilde{K}_{Pi}| > \frac{|K_{yi}|}{\alpha_i} + \left(|h_{ui}| + \frac{|K_{ui}|}{\alpha_i} \right) v. \quad (16)$$

By increasing the observer gain α_i , the size of the ultimate error can be guaranteed.

In this paper, we employ an adaptive approach to obtain the human's intention reference trajectory [19]

$$x_r(t + \Delta t) = x_r(t) - Lz(t) \quad (17)$$

where t denotes the current time in the adaption and Δt denotes the constant time interval during the adaption. At the initial time t_0 , the trajectory is initialized as the desired trajectory, i.e., $x_r(t_0) = x_d(t_0)$. L is a constant matrix that would make the reference trajectory convergent and the item $z(t)$ is defined as follows:

$$z = (\dot{x} - \dot{x}_d) + \Lambda(x - x_d) - f_e \quad (18)$$

where z , $\dot{x}$, and $\dot{x}_d$ are the functions of time t , z is the combination of the position and velocity errors with Λ being the weight of the two items while performing tracking, and f_e is the filtered force, defined in (22). Then, we have

$$\dot{z} = (\ddot{x} - \ddot{x}_d) + \dot{\Lambda}(x - x_d) + \Lambda(\dot{x} - \dot{x}_d) - \dot{f}_e \quad (19)$$

such that

$$\begin{aligned}w = \dot{z} + \Gamma z &= (\ddot{x} - \ddot{x}_d) + (\Lambda + \Gamma)(\dot{x} - \dot{x}_d) \\ &\quad + (\dot{\Lambda} + \Gamma\Lambda)(x - x_d) - (\dot{f}_e + \Gamma f_e).\end{aligned}\quad (20)$$

Compared with (7), we have coefficients in the following form:

$$\begin{cases} K_D = \Gamma + \Lambda \\ K_P = \dot{\Lambda} + \Gamma\Lambda \end{cases} \quad (21)$$

$$\dot{f}_e + \Gamma f_e = K_F F_x. \quad (22)$$

The coefficient K_P is obtained by the stiffness observer and updated by (12), and the coefficient K_D is obtained by the relation between the stiffness and the damper, $K_{Di} = \sqrt{2K_{Pi}}$. The inertia coefficient is set to be an identity matrix in this paper. In summary, we have the following theorem for the adaption of the reference trajectory.

Theorem 1: Considering the human–robot interaction dynamics (1) that satisfies Assumption (1) and Assumption (2),

using the impedance model (6) and the trajectory adaptation algorithm (17) under the control proposed later in (36), the objective (7) can be minimized to 0.

The proof can be found in Appendix A.

B. Control Design

The overall control framework is shown in Fig. 2. In the two-loop control scheme, the reference trajectory is generated according to the impedance model in the outer loop. In the inner loop, the adaptive control is proposed to deal with the unknown dynamics of the robot.

Assume that full state information q and $\dot{q}$ are available. According to (1), let $x_1 = [q_1, q_2, q_3, \dots, q_n]^T$, $x_2 = [\dot{q}_1, \dot{q}_2, \dot{q}_3, \dots, \dot{q}_n]^T$, then the dynamics of the interaction task can be written in the following form:

$$\dot{x}_1 = x_2 \quad (23)$$

$$\dot{x}_2 = M^{-1}[\tau + \tau_e - f_{\text{dis}} - G - Cx_2]. \quad (24)$$

Consider discrete points of the reference trajectory (17), a continuous trajectory is needed to be fitted online. Here, Bezier curve is employed. The definition of a parametric Bezier curve can be expressed as follows:

$$Q(u) = \sum_{i=0}^p A_i J_{p,i}(u), \quad 0 \leq u \leq 1 \quad (25)$$

where u is a normalized parameter, p is the degree of the curve, and A_i is the i th control point of the Bezier curve. The i th Bezier function is $J_{p,i}(u) = {}^p C_i u^i (1-u)^{p-i}$, and $u^i (1-u)^{p-i}$ is the blending function, where ${}^p C_i = \frac{p!}{(p-i)!i!}$. Here, a three-degree Bezier curve defined by four control points is used to achieve the continuous trajectory

$$\begin{aligned}x_p(u) &= A_0(1-u)^3 + 3A_1u(1-u)^2 \\ &\quad + 3A_2u^2(1-u) + A_3u^3 \\ &= a_0 + a_1u + a_2u^2 + a_3u^3\end{aligned}\quad (26)$$

where A_0, A_1, A_2, A_3 are control points and a_0, a_1, a_2, a_3 are the corresponding coefficients. According to [20], the radius of curvature varies smoothly in this Bezier curve for its high-order differential, which is existing, so that we have the continuous reference trajectory $x_r^* = x_p(u)$. The errors z_1 and z_2 are defined as follows:

$$z_1 = x_1 - x_r^* \quad (27)$$

$$z_2 = x_2 - \alpha_1 \quad (28)$$

where α_1 is the virtual control to z_1 . Considering an n -degrees of freedom robotic exoskeleton, $\alpha_1 \in \mathbb{R}^n$, $z_1 \in \mathbb{R}^n$, and $z_2 \in \mathbb{R}^n$, we have

$$\dot{z}_1 = \dot{x}_1 - \dot{x}_r^* = z_2 + \alpha_1 - \dot{x}_r. \quad (29)$$

Considering a Lyapunov function candidate $V_1 = \frac{1}{2} z_1^T z_1$, the time derivative of V_1 is

$$\dot{V}_1 = z_1^T \dot{z}_1 = z_1^T (z_2 + \alpha_1 - \dot{x}_r). \quad (30)$$

Let $\alpha_1 = \dot{x}_r^* - K_1 z_1$ with $K_1 \in \mathbb{R}^{n \times n}$ and $\lambda_{\min}(K_1) > 0$, the Lyapunov function can be written in the following form:

$$\dot{V}_1 = -z_1^T K_1 z_1 + z_1^T z_2 \quad (31)$$

then, we can have

$$\begin{aligned} \dot{z}_2 &= \dot{x}_2 - \dot{\alpha}_1 \\ &= M^{-1}[\tau + \tau_e(t) - f_{\text{dis}} - G - Cx_2] - \dot{\alpha}_1 \end{aligned} \quad (32)$$

where $\dot{\alpha}_1 = -K_1 \dot{z}_1 + \ddot{x}_r^*$. We consider a Lyapunov function $V_2 = V_1 + \frac{1}{2} z_2^T M z_2$, so the time derivative of V_2 will be

$$\begin{aligned} \dot{V}_2 &= \dot{V}_1 + z_2^T M \dot{z}_2 + \frac{1}{2} z_2^T \dot{M} z_2 \\ &= -z_1^T K_1 z_1 + z_1^T z_2 + z_2^T \left(\tau + \tau_e - f_{\text{dis}} \right. \\ &\quad \left. - G - Cx_2 - M\dot{\alpha}_1 + \frac{1}{2} \dot{M} z_2 \right). \end{aligned} \quad (33)$$

Applying Property 1, we have

$$\begin{aligned} \dot{V}_2 &= -z_1^T K_1 z_1 + z_1^T z_2 + z_2^T (\tau + \tau_e - f_{\text{dis}} \\ &\quad - G - C\alpha_1 - M\dot{\alpha}_1). \end{aligned} \quad (34)$$

Once if the parameters of the dynamics are all known, a control method is expressed in the following form:

$$\tau = -z_1 - K_2 z_2 + f_{\text{dis}} + G + C\alpha_1 + M\dot{\alpha}_1 - \tau_e. \quad (35)$$

Nevertheless, it is not easy to have the precise information of disturbance f_{dis} as well as the terms of the robotic dynamics including G, C, M . Property 3 is applied to deal with the unknown dynamics. Moreover, the external disturbance is compensated by a disturbance observer. The adaptive control law is given as

$$\tau = -z_1 - K_2 z_2 + Y(Z)\hat{W} + \hat{f} - \tau_e \quad (36)$$

with $K_2 \in \mathbb{R}^{n \times n}$ and $\lambda_{\min}(K_2) > 0$, and $\hat{f}$ is the high-order disturbance observer, and the disturbance observer is given in the following form [25]:

$$\begin{cases} \dot{\hat{f}} = \hat{z} - K_d x_2 \\ \dot{\hat{z}} = K_d (Y_d(Z_d)\hat{W}_d - \hat{f}) \end{cases} \quad (37)$$

where $Y_d(Z_d) \in \mathbb{R}^{n \times m}$ is the dynamic regressor matrix, $Z = [x_1^T, x_2^T, \alpha_1^T, \dot{\alpha}_1^T]$, $Z_d = [\tau^T, x_1^T, x_2^T]$, and $K_d^T = K_d > 0$. $W \in \mathbb{R}^m$ and $W_d \in \mathbb{R}^m$ are physical parameters and their updating laws are designed as

$$\dot{W}_i = -\Gamma_i (Y_i(Z) z_{2i} + \theta_i \hat{W}_i) \quad (i = 1, 2, \dots, m) \quad (38)$$

$$\dot{W}_{di} = -\Gamma_{di} (\theta_{di} \hat{W}_{di}) \quad (i = 1, 2, \dots, m) \quad (39)$$

where θ_i and θ_{di} are small positive real numbers, $\Gamma_i > 0$, and $\Gamma_{di} > 0$. The characteristic of the regressor is also used in the disturbance observer

$$Y(Z)W^* = G(x_1) + C(x_1, x_2)\alpha_1 + M\dot{\alpha}_1 - \epsilon \quad (40)$$

$$Y_d(Z_d)W_d^* = M^{-1}(\tau + \tau_e - C(x_1, x_2)x_2 - G(x_1)) - \epsilon_d \quad (41)$$



Fig. 3. Developed exoskeleton system.

where W^* and W_d^* are optimal values, ϵ and ϵ_d are estimate errors, ϵ and ϵ_d satisfy $\max_{Z \in \Omega_Z} |\epsilon| < \epsilon^*$, and $\max_{Z_d \in \Omega_{Z_d}} |\epsilon_d| < \epsilon_d^*$, respectively [23].

Remark 1: In the adaptive control law (36), the parameters are uncertain and there may exist deviations, but the control (36) can maintain its robustness once the parameters are chosen in the bounded region, i.e., $\theta_i > 0$, $\theta_{di} > 0$, $K_1 = K_1^T > 0$ and $K_2 - 1.5I_{n \times n} = (K_2 - 1.5I_{n \times n})^T > 0$ and $K_d - 2I_{n \times n} = (K_d - 2I_{n \times n})^T > 0$, which is proved in Appendix B. With such parameters, the robustness of the proposed control is guaranteed.

Theorem 2: Consider the robotic dynamics (23) and (24), using (36), together with (37), and the adaptive laws (38) and (39), and the overall control scheme is shown in Fig. 2, the control signals of the closed-loop system, z_1 , z_2 , $\tilde{W}$, and $\tilde{W}_d$ are semiglobally bounded. Furthermore, the error signals z_1 , z_2 , $\tilde{W}$, and $\tilde{W}_d$ will be kept in the compact sets Ω_{z1} , Ω_{z2} , $\Omega_{\tilde{W}}$, and $\Omega_{\tilde{W}_d}$ respectively, defined as follows: $\Omega_{z1} = \{z_1 \in \mathbb{R}^n \mid \|z_{1i}\| \leq \sqrt{D_1}\}$, $\Omega_{z2} = \{z_2 \in \mathbb{R}^n \mid \|z_{2i}\| \leq \sqrt{\frac{D_1}{\lambda_{\min}(M)}}\}$, $\Omega_{\tilde{W}} = \{\tilde{W} \in \mathbb{R}^m \mid \|\tilde{W}_i\| \leq \sqrt{\frac{D_1}{\lambda_{\min}(\Gamma^{-1})}}\}$, $\Omega_{\tilde{W}_d} = \{\tilde{W}_d \in \mathbb{R}^m \mid \|\tilde{W}_{dk}\| \leq \sqrt{\frac{D_1}{\lambda_{\min}(\Gamma_d^{-1})}}\}$, where $D_1 = 2(V_2^*(0) + \frac{B_1}{\kappa_1})$ with κ_1 and B_1 given in (70) and (71), where both are positive definite.

The proof can be found in Appendix B.

IV. EXPERIMENTS

The experiments for the proposed adaptive control schemes are designed to verify its effectiveness, which has been conducted by the developed robotic exoskeleton located in South China University of Technology (SCUT) laboratory shown in Fig. 3. Two dc motors are equipped for the exoskeleton robot system as actuators and the selection of the actuators is on the basis of our actual needed torque in experiments and the external force in the tasks. DC motors are chosen as Maxon EC90Flat, and the motor drivers are chosen as Elmo SOL-WHI5/60E01. The motor drivers are connected with the industrial personal computer (IPC) using CANopen Communication Module. The maximum baud rate of the CAN bus is 1 Mb/s. The maximum sampling rate of the force sensor is 1 kHz in the outer loop and the sampling rate of the control loop is 123 Hz in the inner loop. Fig. 3 presents the experimental platform, which contains the robotic system with a force sensor, a unit of the execu-

TABLE I
INFORMATION OF THE SUBJECTS

Experiment Number	Subject Number	Age	Weight	Results
1	1	24	54.5 kg	Fig. 4
2	2	23	56.5 kg	Fig. 5
3	3	22	62.1 kg	Fig. 6

tive drivers, and an industrial personal computer. The executive drivers are used to generate a driving torque for the actuators and gather the motion information in the tasks from the force sensor and encoders of actuators, while the IPC is used to execute the programs that involve the control on the experiment platform.

The gains of the controller are chosen as $K_1 = \text{diag}[24.5, 30.6]$, $K_2 = \text{diag}[3.3, 2.1]$, $K_d = \text{diag}[2.2, 2.8]$. The gains Γ_i and $\Gamma_{d,i}$ are defined as $\Gamma_1 = 0.01I$, $\Gamma_2 = 0.01I$, $\Gamma_{d,1} = 0.04I$, and $\Gamma_{d,2} = 0.04I$. Small positive constants θ_i and $\theta_{d,i}$ are chosen as $\theta_1 = 0.5$, $\theta_2 = 0.5$, $\theta_{d,1} = 0.1$, and $\theta_{d,2} = 0.1$.

Three experimental subjects participated in the experiment. Table I shows the relevant information of the three subjects. During the task of the experiment, the subject holds the end-effector and moves according to his own intention back and forth in an appropriate range, which can be expressed by the shaped reference trajectory. The interaction force is generated and measured by the force sensor equipped at the end of the robotic exoskeleton. With the adaptive control method proposed in this paper, the interaction force will decrease and the errors of tracking the reference trajectory are convergent.

The results are shown in Figs. 4–8. Figs. 4(a) and (b), 5(a) and (b), and 6(a) and (b) show the tracking situation of the shoulder joint (q_1) and the elbow joint (q_2) of the robot. Figs. 4(c), 5(c), and 6(c) show the trajectories in task space. Figs. 4(d), 5(d), and 6(d) show the output of the stiffness observer. The interaction forces measured by the force sensor are shown in Fig. 8, and the interaction torques are shown in Figs. 4(g), 5(g), and 6(g), respectively. From these figures, we can see that the interaction forces and torques are decreasing such that the task can be performed easily and effectively.

The velocities of the two robot joints are shown in Figs. 4(e), 5(e), and 6(e). The input torques of the robot joints are shown in Figs. 4(h), 5(h), and 6(h). In order to analyze the qualities of the manipulability, the force ellipsoids of the exoskeleton are presented in Figs. 4(f), 5(f), and 6(f). As is mentioned in [26], according to the Jacobian in (4), taking the singular value decomposition of J , we have $J = U\Sigma V^T$, where $U = [u_1, u_2] \in R^{2 \times 2}$ and $V \in R^{2 \times 2}$ are orthogonal matrices, $\Sigma = \text{diag}[\sigma_1, \sigma_2]$ is a diagonal matrix of singular values. The principal axes are given by $(1/\sigma_1)u_1$ and $(1/\sigma_2)u_2$. The manipulability measure is defined as $w = |\det J| = L_1 L_2 |\sin(q_2)|$. Therefore, if the angle of the elbow joint $q_2 = \frac{\pi}{2}$, the exoskeleton takes its best posture.

In [27]–[31], admittance and impedance control were investigated to deal with the human–robot interaction. In [27], an admittance control scheme was proposed in the human–robot

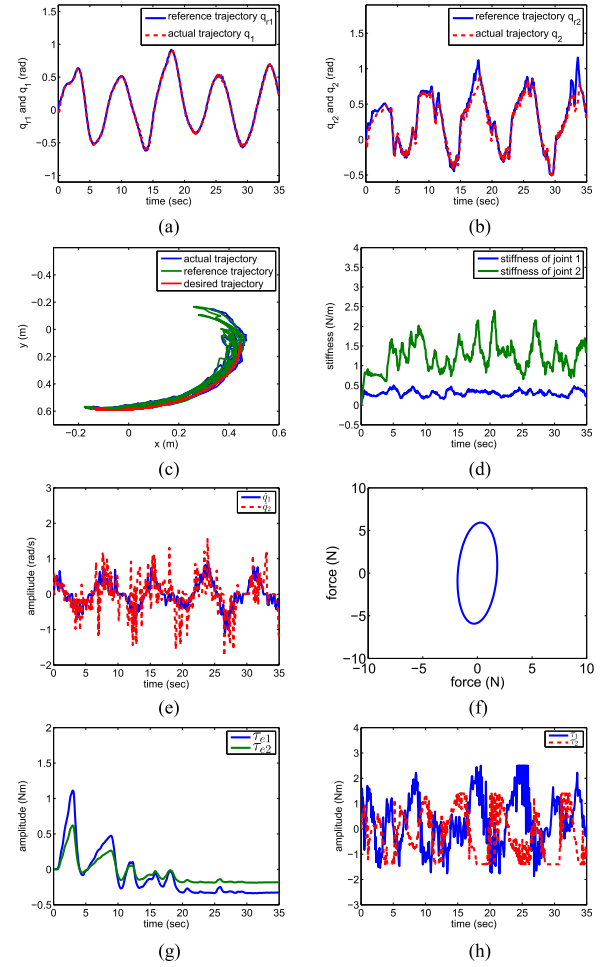


Fig. 4. Experimental results of subject 1. (a) Tracking trajectory of the shoulder joint. (b) Tracking trajectory of the elbow joint. (c) Trajectory in task space. (d) Stiffness. (e) Velocity measurement of two joints. (f) Force ellipsoid of subject 1. (g) F_e . (h) Input torques of the joints.

cooperation tasks with online adaptation of the admittance control gains. However, the scheme did not consider the shaping of the reference trajectory or the model matching error. In [28], a two-loop admittance control formulation was developed to assist humans with varying levels of skill to achieve task-specific objectives, where the inner loop guaranteed robustness and stability of the motion control and the outer loop tuned the admittance model. However, the mass, damping, and stiffness parameters were not designed in the control scheme, such that the impedance information of the human–robot system might not be acquired. In this paper, reference trajectory shaping is considered in (17) and the adaption is able to make the model matching error w in (7) converge to 0. The impedance parameters are obtained by (12) and used to derive the coefficients in (21), such that the outer loop is tuned to make the model matching error converge to 0.

In [29], an interaction control strategy was presented for a gait rehabilitation robot that was driven by a series elastic actuator. However, the strategy did not consider the trajectory shaping and the stiffness coefficient was constant and chosen according to the actuator, such that the parameter might not reflect the

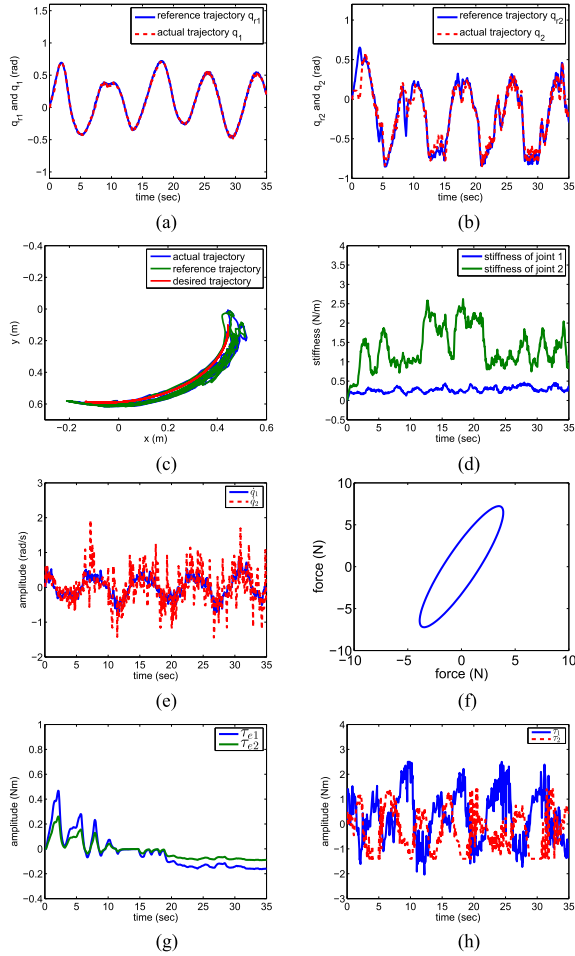


Fig. 5. Experimental results of subject 2. (a) Tracking trajectory of the shoulder joint. (b) Tracking trajectory of the elbow joint. (c) Trajectory in task space. (d) Stiffness. (e) Velocity measurement of two joints. (f) Force ellipsoid of subject 2. (g) τ_e . (h) Input torques of the joints.

varying stiffness of the human subjects. In [30], an impedance control structure was presented for the human–robot interaction, which consisted of a robot-specific inner loop and a task-specific outer loop. The optimal impedance parameters were obtained through reinforcement learning, but the method did not consider the adaption of the reference trajectory. In [31], impedance control was proposed for the human–robot comanipulation to make the robot adaptive to the human motor behavior. The stiffness of the robot was obtained through the electromyography signals of human muscles, but the method did not consider the relationship between force and the robot motion. The work mentioned above did not consider the shaping of the reference trajectory. In this paper, the stiffness is updated according to the force and the robot motion in a closed loop (12). The reshaped trajectory (17) is able to make the robot perform according to the desired model (6) such that the robot motion is compliant and the comanipulation would be effective.

In order to compare between the proposed method and the method that uses constant impedance, a group of comparative experiment has been conducted. The results of the comparative experiment are shown in Fig. 7. In the comparative ex-

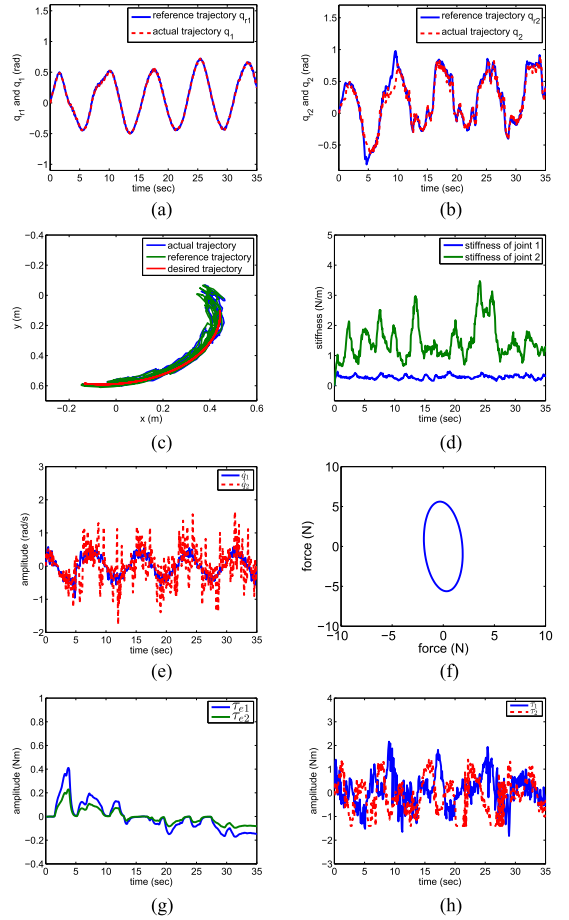


Fig. 6. Experimental results of subject 3. (a) Tracking trajectory of the shoulder joint. (b) Tracking trajectory of the elbow joint. (c) Trajectory in task space. (d) Stiffness. (e) Velocity measurement of two joints. (f) Force ellipsoid of subject 3. (g) τ_e . (h) Input torques of the joints.

periment, the impedance parameters are chosen as constants: $K_F = \text{diag}[1, 1]$, $K_D = \text{diag}[0.9, 1.5]$, $K_P = \text{diag}[0.4, 1.2]$. The interaction forces are shown in Fig. 8.

In the conventional method used in the comparative experiment, the adaption of the stiffness (12) is not used and the impedance parameters are chosen as constants. Thus, the coefficients calculated by the stiffness through (21) might not make the reference trajectory compliant to the interaction force. That is to say, the interaction force would not decrease as is shown in Fig. 8. In the proposed adaptive admittance control, to solve the cost function (5), the reference trajectory shaping (17) and the stiffness adaption (12) are both presented to make the robot motion compliant to the interaction force exerted by the human subjects. Thus, the interaction force would be minimized comparing to the conventional method using constants impedance parameters, which can be seen in Fig. 8.

In summary, the novelties and effectiveness of the proposed method can be clarified as follows. (i) The reference trajectories of the robot are reshaped according to the interaction force exerted by the human subjects, which is verified by the experiments 1–3 in Figs. 4(a)–(c), 5(a)–(c), and 6(a)–(c). Figs. 4(a), 5(a), and 6(a) show the reference trajectories and the actual trajectories of the shoulder joint in the experiments 1–3, respec-

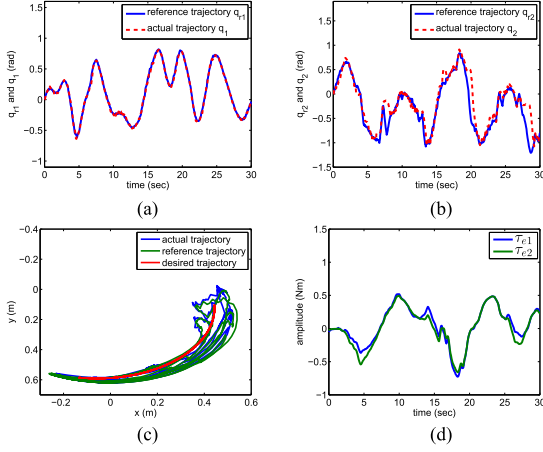


Fig. 7. Comparative experimental result. (a) Tracking trajectory of the shoulder joint. (b) Tracking trajectory of the elbow joint. (c) Trajectory in task space. (d) τ_e .

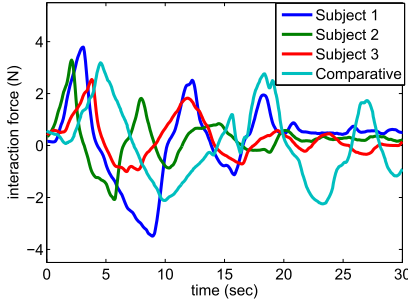


Fig. 8. Interaction force.

tively. Similarly, Figs. 4(b), 5(b), and 6(b) show the trajectories of the elbow joint. Figs. 4(c), 5(c), and 6(c) show the trajectories in task space. We can see that the trajectories changed according to the interaction force so that the motion of the robot was compliant with the human subjects. (ii) The impedance parameters can be obtained through the proposed estimated stiffness (12) based on the varying stiffness of the human subjects, which has been verified by the experiments 1–3 in Figs. 4(d), 5(d), and 6(d); those figures show the time-varying stiffness of the three human subjects in the experiments 1–3, respectively, and indicate the impedance parameters varying in real time, and the outer loop control has been tuned to make the robot perform according to the desired impedance model. (iii) The interaction force in the proposed method was smaller comparing with the conventional method used in the comparative experiment, which has been verified by the experiments 1–4 in Fig. 8 showing the interaction force exerted by the human subjects, which indicates that the robot is more compliant to human operation with small human effort when using the proposed control.

The estimation accuracy of the interaction force can be explained according to (14) and (15). The information of the interaction force is included in the update law of the stiffness (12), which is feedback in a closed loop. Using the stiffness update law, the errors of the stiffness and the interaction

force would converge with uniformly ultimately bounded errors, which are illustrated in (14) and (15). Thus, the estimation accuracy of the interaction force can be guaranteed.

Under the adaptive admittance control proposed in this paper, the interaction force is gradually reduced synchronously with the adaption of the stiffness estimated by the observer. The reduction of the interaction force can demonstrate that the robotic exoskeleton is showing compliance to the human behavior during the task. In summary, from these results we can see that our proposed control approach is effective in the physical interaction between human and robot.

V. CONCLUSION

This paper presented a framework of adaptive admittance control by incorporating the human's motion intention, which consists of double control loops. The inner loop was used to deal with the unknown masses and moments of inertia of the robot dynamics in a feedback way, while the outer loop was used to tune the interaction model by considering the motion intention of human subjects. The adaptive technique has been utilized to deal with the unknown dynamics in the inner loop such that the effect of the task performed in the outer loop can be ensured. The experimental results show the better performance of the proposed admittance control in physical interaction tasks between humans and robots. Without prior offline model tuning, the proposed control approach can also be robust even if the robotic dynamics are unknown beforehand.

APPENDIX A PROOF OF THEOREM 1

Lemma 2 (see [32]): Considering the signals $g(t)$, $\alpha(t)$, and $h(t)$ satisfying the following condition:

$$g(t) \leq \alpha(t) + \int_0^t h(\delta)g(\delta)d\delta \quad (42)$$

then we have

$$g(t) \leq \alpha(t) + \int_0^t \alpha(\delta)h(\delta)e^{\int_\delta^t h(s)ds}d\delta. \quad (43)$$

Combining the control law (36), and z_1 (27), z_2 (28), we can change the control τ into the following form:

$$\tau = -K(x - x_r) - K_2(\dot{x} - \dot{x}_r) + Y(Z)\hat{W} + \hat{f} - \tau_e \quad (44)$$

where $K = K_2\Gamma$ is the stiffness coefficient and the control is function of time t . We can define as

$$s_r(t) = Kx_r(t) + K_2\dot{x}_r(t) \quad (45)$$

$$s(t) = Kx(t) + K_2\dot{x}(t) \quad (46)$$

where $\dot{x}_r(t) \approx \frac{1}{\Delta t}(x_r(t) - x_r(t - \Delta t))$. Consider the time $t + \Delta t$ in (45), we have $s_r(t + \Delta t) = Kx_r(t + \Delta t) + K_2\dot{x}_r(t + \Delta t)$. From (17) and (20), substitute $x_r(t + \Delta t) = x_r(t) -$

$Lz(t)$ and $w = \dot{z} + \Gamma z$ into $s_r(t + \Delta t)$, one can obtain

$$\begin{aligned} s_r(t + \Delta t) &= Kx_r(t + \Delta t) + K_2\dot{x}_r(t + \Delta t) \\ &= K(x_r(t) - Lz(t)) + K_2(\dot{x}_r(t) - L\dot{z}(t)) \\ &= Kx_r(t) + K_2\dot{x}_r(t) - (KLz(t) + K_2L\dot{z}(t)) \\ &= s_r(t) - K_2Lw(t). \end{aligned} \quad (47)$$

Define $x = x_e$ as the trajectory that satisfies the following equation:

$$(\ddot{x}_e - \ddot{x}_d) + K_D(\dot{x}_e - \dot{x}_d) + K_P(x_e - x_d) = K_F F_e. \quad (48)$$

Then, we have $w = 0$ and F_e is the external force when $x = x_e$. The actual trajectory is initialized at the initial time: $x(t_0) = x_d(t_0)$. Substituting control (44) into the interaction dynamics, we can have

$$M(x)\ddot{x} + C\dot{x} + G + f_{\text{dis}} = -s + s_r + Y\hat{W} + \hat{f} \quad (49)$$

With the trajectory x_e , we can also define the reference trajectory x_{re} that satisfies

$$M(x_e)\ddot{x}_e + C\dot{x}_e + G + f_{\text{dis}} = -s_e + s_{re} + Y\hat{W} + \hat{f} \quad (50)$$

where $s_e = Kx_e + K_2\dot{x}_e$ and $s_{re} = Kx_{re} + K_2\dot{x}_{re}$ deriving from (45), and $w = 0$ when $x = x_{re}$.

Considering $N(x, \dot{x}) = C\dot{x} + G + f_{\text{dis}} - Y\hat{W} - \hat{f} + s$, then (49) and (50) can be written as follows:

$$M(x)\ddot{x} + N(x, \dot{x}) = s_r \quad (51)$$

$$M(x_e)\ddot{x}_e + N(x_e, \dot{x}_e) = s_{re}. \quad (52)$$

Defining $\Delta M = M(x) - M(x_e)$, $\Delta N = N(x, \dot{x}) - N(x_e, \dot{x}_e)$, $\Delta x = x - x_e$, $\Delta s_r = s_r - s_{re}$, $\Delta F = F_x - F_e$. Combining (51), (52), (48), and (20), we have

$$\Delta\ddot{x} = M^{-1}(x)(\Delta s_r - \Delta N - \Delta M\ddot{x}_e) \quad (53)$$

$$w = \Delta\ddot{x} + K_D\Delta\dot{x} + K_P\Delta x - K_F\Delta F \quad (54)$$

where $\Delta\ddot{x}$ and w are functions about the time t . From (48), (54), and the definition that $\Delta s_r(t + \Delta t) = s_r(t + \Delta t) - s_r(t)$, $\Delta s_r(t) = s_r(t) - s_r(t - \Delta t)$, we have

$$\begin{aligned} \Delta s_r(t + \Delta t) &= -K_2L(\Delta\ddot{x} + K_D\Delta\dot{x} \\ &\quad + K_P\Delta x - K_F\Delta F). \end{aligned} \quad (55)$$

We consider $n_L, n_{M1}, n_K, n_D, n_P, n_F, n_2$ as the $\|\cdot\|_\infty$ norm of $L, M^{-1}, K_2, K_D, K_P, K_F, \ddot{x}_e$, respectively, in the compact set of finite time. The function $N(x, \dot{x})$ is continuously derivable so it satisfies the Lipschitz condition. We have Lipschitz coefficients l_n, l_m, l_f of $N(x, \dot{x})$, $M(x)$ and F_x . From (53) and (55), take norm of both sides and there is a constant u_0 that

yields

$$\|\Delta\ddot{x}\| \leq n_{M1}\|\Delta s_r\| + u_0\|[\Delta x, \Delta\dot{x}]\| \quad (56)$$

$$\begin{aligned} \|\Delta s_r(t + \Delta t)\| &\leq \|K_2LM^{-1}\|\|\Delta s_r(t)\| \\ &\quad + \|K_2L\|\left(\|M^{-1}\|\|\Delta N\| \right. \\ &\quad + \|M^{-1}\ddot{x}_e\|\|\Delta M\| + \|K_D\|\|\Delta\dot{x}\| \\ &\quad + \|K_P\|\|\Delta x\| + \|K_F\|\|\Delta F\|\left.\right) \\ &\leq q\|\Delta s_r(t)\| + u_1\|[\Delta x, \Delta\dot{x}]\|. \end{aligned} \quad (57)$$

Note that $q = \|K_2LM^{-1}\| < 1$ and $u_1 = n_K n_L (n_{M1} l_n + n_{M1} n_2 l_m + n_D + n_P + n_F l_f)$. Using integral and the consideration of $\Delta x(0) = 0$, then we have

$$\begin{aligned} \|[\Delta x(t), \Delta\dot{x}(t)]\| &\leq \int_0^t \left((u_0 + 1)\|[\Delta x(\delta), \Delta\dot{x}(\delta)]\| \right. \\ &\quad \left. + n_{M1}\|\Delta s_r(\delta)\| \right) d\delta. \end{aligned} \quad (58)$$

From Lemma 2, multiply both sides by $e^{-\alpha t}$, such that

$$e^{-\alpha t}\|[\Delta x, \Delta\dot{x}]\| \leq n_{M1} \int_0^t e^{-\alpha\delta} \|\Delta s_r\| e^{(u_0+1-\alpha)(t-\delta)} d\delta. \quad (59)$$

As is shown in [33], $\forall M(t), \|M(t)\|_\alpha = \sup(e^{-\alpha t}\|M(t)\|)$, then the following inequation holds that $\|M(t)\|_\alpha \leq \|M(t)\|_\infty \leq e^{\alpha t}\|M(t)\|_\alpha$. So the above inequation (59) can be written as follows:

$$\begin{aligned} \|[\Delta x, \Delta\dot{x}]\|_\alpha &\leq n_{M1}\|\Delta s_r\|_\alpha \int_0^t e^{(u_0+1-\alpha)(t-\delta)} d\delta \\ &\leq q_1\|\Delta s_r\|_\alpha. \end{aligned} \quad (60)$$

Note that $q_1 = \left\| \frac{n_{M1}(1-e^{(u_0+1-\alpha)T})}{\alpha-u_0-1} \right\|$, where T is the finite time interval. Combining (56), we have

$$\|\Delta\ddot{x}\|_\alpha \leq q_2\|\Delta s_r\|_\alpha \quad (61)$$

where $q_2 = n_{M1} + u_0 q_1$. From (54), we have

$$\begin{aligned} \|w\|_\alpha &\leq \|\Delta\ddot{x}\|_\alpha + u_2\|[\Delta x, \Delta\dot{x}]\|_\alpha \\ &\leq q_3\|\Delta s_r\|_\alpha \end{aligned} \quad (62)$$

where $u_2 = n_D + n_P + n_F l_F$, $q_3 = q_2 + u_2 q_1$. Combining (57) and (60), we have

$$\begin{aligned} \|\Delta s_r(t + \Delta t)\|_\alpha &\leq q\|\Delta s_r(t)\|_\alpha + u_1 q_1\|\Delta s_r(t)\|_\alpha \\ &\leq q_4\|\Delta s_r(t)\|_\alpha \end{aligned} \quad (63)$$

where $q_4 = q + \frac{u_1 n_{M1}(1-e^{(u_0+1-\alpha)T})}{\alpha-u_0-1}$. Here, if we make α large enough, then we get $q_4 < 1$, $\|\Delta s_r\|_\alpha \rightarrow 0$, such that $\|\Delta s_r\| \rightarrow 0$. Then from (62), $w \rightarrow 0$ is guaranteed.

APPENDIX B PROOF OF THEOREM 2

Substituting the control (36) into (34)

$$\begin{aligned} \dot{V}_2 = & -z_1^T K_1 z_1 - z_2^T K_2 z_2 - z_2^T \epsilon \\ & - z_2^T f_{\text{dis}} + z_2^T \hat{f} + z_2^T Y(Z) \tilde{W} \end{aligned} \quad (64)$$

where $Z = [x_1^T, x_2^T, \alpha_1^T, \dot{\alpha}_1^T]$ and $Z_d = [\tau^T, x_1^T, x_2^T]$. Given $\tilde{W} = \hat{W} - W^*$ and $\tilde{W}_d = \hat{W}_d - W_d^*$, by considering the effect of $\tilde{W}$ to the stability of the control system, the following Lyapunov candidate is selected as follows:

$$\begin{aligned} V_2^* = & V_2 + \frac{1}{2} \sum_{i=1}^n \tilde{W}_i^T \Gamma_i^{-1} \tilde{W}_i + \frac{1}{2} \tilde{f}^T \tilde{f} \\ & + \frac{1}{2} \sum_{k=1}^n \tilde{W}_{dk}^T \Gamma_{dk}^{-1} \tilde{W}_{dk} \end{aligned} \quad (65)$$

where $\tilde{f} = \hat{f} - f_{\text{dis}}$. Derivative of V_2^* , we have

$$\begin{aligned} \dot{V}_2^* = & \sum_{i=1}^n (z_{2,i} Y_i(Z) \tilde{W}_i + \tilde{W}_i^T \Gamma_i^{-1} \dot{\tilde{W}}_i) \\ & + \sum_{k=1}^n \tilde{W}_{dk}^T \Gamma_{dk}^{-1} \dot{\tilde{W}}_{dk} - z_1^T K_1 z_1 - z_2^T K_2 z_2 \\ & - z_2^T \epsilon - z_2^T f_{\text{dis}} + \tilde{f}^T \dot{\tilde{f}} + z_2^T \hat{f}. \end{aligned} \quad (66)$$

Consider the disturbance observer (37), we have

$$\dot{\tilde{f}} = -K_d \tilde{f} - K_d (M^{-1} + 1) f_{\text{dis}} - \dot{f}_{\text{dis}} + K_d \epsilon_d. \quad (67)$$

Let $K_a = K_d (M^{-1} + 1)$. With the disturbance observer error signals (67), and the updating laws (38) and (39), we have

$$\begin{aligned} \dot{V}_2^* = & -z_1^T K_1 z_1 - z_2^T K_2 z_2 - z_2^T \epsilon + z_2^T \tilde{f} - \tilde{f}^T K_d \tilde{f} \\ & - \tilde{f}^T \dot{\tilde{f}}_{\text{dis}} - \tilde{f}^T K_a f_{\text{dis}} + \tilde{f}^T K_d \epsilon_d - \sum_{i=1}^n \theta_i \tilde{W}_i^T \hat{W}_i \\ & - \sum_{k=1}^n \theta_{dk} \tilde{W}_{dk}^T \hat{W}_{dk}. \end{aligned} \quad (68)$$

Since $-\tilde{W}_i^T \hat{W}_i = -\tilde{W}_i^T (W_i^* + \tilde{W}_i) = -\tilde{W}_i^T \tilde{W}_i - \tilde{W}_i^T W_i^*$ and $-\tilde{W}_i^T W_i^* \leq \frac{1}{2} (\tilde{W}_i^T \tilde{W}_i + W_i^{*T} W_i^*)$, we have $-\tilde{W}_i^T \hat{W}_i \leq -\frac{1}{2} \tilde{W}_i^T \tilde{W}_i + \frac{1}{2} W_i^{*T} W_i^*$. Similarly, $-\tilde{W}_{di}^T \hat{W}_{di} \leq -\frac{1}{2} \tilde{W}_{di}^T \tilde{W}_{di} + \frac{1}{2} W_{di}^{*T} W_{di}^*$. And $z_2^T \epsilon \leq \frac{1}{2} z_2^T z_2 + \frac{1}{2} \epsilon^T \epsilon \leq \frac{1}{2} z_2^T z_2 + \frac{1}{2} \|\epsilon^*\|^2$, $z_2^T \tilde{f} \leq \frac{1}{2} z_2^T z_2 + \frac{1}{2} \tilde{f}^T \tilde{f}$, $-\tilde{f}^T \dot{\tilde{f}}_{\text{dis}} \leq \frac{1}{2} \tilde{f}^T \tilde{f} + \frac{1}{2} \|f^*\|^2$, $-\tilde{f}^T K_a f_{\text{dis}} \leq \frac{1}{2} \tilde{f}^T \tilde{f} + \frac{1}{2} \|K_a\|^2 \|f_M^*\|^2$, $-\tilde{f}^T K_d \epsilon_d \leq \frac{1}{2} \tilde{f}^T \tilde{f} + \frac{1}{2} \|K_d\|^2 \|\epsilon_d^*\|^2$. Substituting these inequalities and $e_f = \hat{f} - f_{\text{dis}}$ into (68),

one has

$$\begin{aligned} \dot{V}_2^* \leq & -z_1^T K_1 z_1 - z_2^T (K_2 - 1.5 I_{n \times n}) z_2 - \tilde{f}^T (K_d - 2I) \tilde{f} \\ & - \sum_{i=1}^n \frac{1}{2} \theta_i \tilde{W}_i^T \tilde{W}_i - \sum_{k=1}^n \frac{1}{2} \theta_{dk} \tilde{W}_{dk}^T \tilde{W}_{dk} + \frac{1}{2} \|\epsilon^*\|^2 \\ & + \frac{1}{2} \|f^*\|^2 + \frac{1}{2} \|K_d\|^2 \|\epsilon_d^*\|^2 + \frac{1}{2} \|K_a\|^2 \|f_M^*\|^2 \\ & + \sum_{i=1}^n \frac{1}{2} \theta_i W_i^{*T} W_i^* + \sum_{k=1}^n \frac{1}{2} \theta_{dk} W_{dk}^{*T} W_{dk}^* \\ \leq & -\kappa_1 V_2^* + B_1 \end{aligned} \quad (69)$$

where

$$\begin{aligned} \kappa_1 = & \min \left(2\lambda_{\min}(K_1), 2\lambda_{\min}(K_d - 2I), \frac{2\lambda_{\min}(K_2 - 1.5 I_{n \times n})}{\lambda_{\max}(M)}, \right. \\ & \left. \min_{i=1,2,\dots,m} \left\{ \frac{\theta_i}{\lambda_{\max}(\Gamma_i^{-1})} \right\}, \min_{k=1,2,\dots,m} \left\{ \frac{\theta_{dk}}{\lambda_{\max}(\Gamma_{dk}^{-1})} \right\} \right) \end{aligned} \quad (70)$$

$$\begin{aligned} B_1 = & \frac{1}{2} \|\epsilon^*\|^2 + \sum_{i=1}^n \frac{1}{2} \theta_i W_i^{*T} W_i^* + \frac{1}{2} \|f^*\|^2 + \frac{1}{2} \|K_d\|^2 \|\epsilon_d^*\|^2 \\ & + \frac{1}{2} \|K_a\|^2 \|f_M^*\|^2 + \sum_{k=1}^n \frac{1}{2} \theta_{dk} W_{dk}^{*T} W_{dk}^*. \end{aligned} \quad (71)$$

To ensure $\kappa_1 > 0$, the design parameters $\theta_i > 0$, $\theta_{dk} > 0$, $K_1 = K_1^T > 0$ and $K_2 - 1.5 I_{n \times n} = (K_2 - 1.5 I_{n \times n})^T > 0$ and $K_d - 2 I_{n \times n} = (K_d - 2 I_{n \times n})^T > 0$. Integrating (69) over $[0, t]$, we can get

$$V_2^* \leq \left(V_2^*(0) - \frac{B_1}{\kappa_1} \right) e^{-\kappa_1 t} + \frac{B_1}{\kappa_1} \leq V_2^*(0) + \frac{B_1}{\kappa_1}$$

According to Lemma 1, the system stability is guaranteed.

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